

Leave-One-Out Meta-Analysis

2026-04-17

Introduction

Leave-one-out (LOO) meta-analysis systematically removes each study and re-computes the pooled estimate using the remaining $k - 1$ studies. The resulting per-removal estimates show how much each study drives the overall conclusion. Studies whose removal substantially changes the pooled effect — magnitude, significance, or both — are influential and warrant careful interpretation. LOO is a routine sensitivity analysis in modern meta-analysis reporting.

Prerequisites

A working understanding of pooled meta-analysis (fixed or random effects) and the concept of influence diagnostics from regression analysis.

Theory

For each study $i = 1, \dots, k$, compute $\hat{\theta}_{(-i)}$ by pooling the other $k - 1$ studies under the same meta-analysis model. Plot $\hat{\theta}_{(-i)}$ with its confidence interval across i ; large deviations from the full-model pooled estimate indicate influential studies.

LOO complements other influence measures: Cook's distance and DFBETAS quantify how much each study perturbs the overall fit, and Baujat plots show each study's contribution to heterogeneity (Q) versus its contribution to the pooled estimate. Used together, these diagnostics provide a comprehensive picture of which studies the conclusion depends on.

Assumptions

Same as the underlying random-effects meta-analysis; each leave-one-out fit uses the same model and method as the full pooling.

R Implementation

```
library(metafor)

set.seed(2026)
k <- 10
yi <- c(0.35, 0.42, 0.25, 0.55, 0.28, 0.39, 0.95, 0.31, 0.44, 0.36)
vi <- c(0.02, 0.03, 0.015, 0.02, 0.025, 0.018, 0.01, 0.04, 0.02, 0.022)

res <- rma(yi, vi, method = "REML")
loo <- leavelout(res)

print(loo[, c("estimate", "ci.lb", "ci.ub", "I2")])

# Influence diagnostics: Cook's distance, DFBETAS
inf <- influence(res)
print(inf$inf)
```

Output & Results

`leave1out()` returns a per-study table of the recomputed pooled estimate, confidence interval, and I^2 when each study is omitted. `influence()` provides Cook’s distance, DFBETAS, hat values, and other diagnostics computed jointly.

Interpretation

A reporting sentence: “Leave-one-out analysis revealed study 7 (SMD = 0.95, the largest single estimate) as highly influential; removing it reduced the pooled SMD from 0.41 to 0.35 and shrunk I^2 from 38 % to 18 %. Conclusions are robust to removal of any other study; the sensitivity analysis without study 7 is reported as supplementary.” Always describe what an influential study removal does to both effect size and heterogeneity.

Practical Tips

- Pair LOO with Cook’s distance and DFBETAS from `metafor::influence()`; the combination flags both effect-size influence and overall fit influence.
- If removing any single study crosses the significance line, the conclusion is fragile and should be reported as such.
- LOO is descriptive — it does not correct any bias. Use it to inform readers about robustness, not to “clean up” the meta-analysis post-hoc.
- Report LOO results only if at least one study is markedly influential; otherwise the table conveys no useful information beyond the headline pooled estimate.
- Combine with Baujat plots (`metafor::baujat()`) for a richer visualisation: studies in the upper-right corner contribute disproportionately to both heterogeneity and effect.
- For meta-analyses with fewer than 5 studies, LOO is unstable and largely uninformative; report it but interpret cautiously.

R Packages Used

`metafor::leave1out()` for the canonical LOO output; `metafor::influence()` for Cook’s distance, DFBETAS, and hat values; `metafor::baujat()` for Baujat plots; `meta` for analogous sensitivity tools in an alternative interface; `dmatar::infA()` for combined influence and outlier diagnostics in an integrated workflow.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis